

LAB #4 – Write Simple Script Files

Purpose:

To practice how to write simple script files in MATLAB.

Before the lab:

Read the course slides and chapter 4 of the textbook.

During the lab:

1. With MATLAB, finish the simple problems 1 to 6 in the textbook (section 4.7).
2. In your textbook, look at the more advanced problems after number 20 (section 4.7), select 25 and 27 from those and do them with MATLAB.
3. Close MATLAB.
4. **Tracing Exercise:** What will be displayed exactly after the following m file is executed? (do not use MATLAB for this - indicate space as #), please show your calculation procedures, if any.

```
p = [0:4:20]
q = [100 200 300 400 500 600]
r = p + q
fprintf ('%5.2f\n%5d\n8.3%d\n', p(3), q(4), r(5), q(2));
n1 = 5.45679
n2= 16.5673
n3 = 66
n4 = -87.355657
fprintf ('%8.5f\n', n4);
fprintf ('%4.4f\n', n4);
fprintf ('%8.5f\n', n1);
fprintf ('%-12d\n', n3);
fprintf ('$B%5.0fS\n', n4);
fprintf ('%7.7f\n', n2);
fprintf ('%3.2f\n%2.2f', n2,n4);
fprintf ('%8.4f\n', n1+n3);
```

Lab Report Submission Requirement:

1. Submit Lab #4 based on the given templates on the Blackboard. Please post all the screenshots of the inputs and running outputs of each step on the lab manual.
2. Please submit your report on the blackboard by the due time.